

BMU Firmware V1012 — Digital Output Functions User Guide

Product: ENKO BMU-52S **Firmware:** V1012 **Document scope:** Digital output function codes, trigger logic, and fan control behavior.

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1. Overview

Each of the BMU-52S's 4 digital outputs is assigned exactly one function code, chosen from a **fixed set of 7 (codes 0–6)**. These are pre-built functions defined in firmware — the operator selects one by number but cannot construct a custom alarm combination.

Function categories:

- **User Configured** — not driven by firmware logic; the output follows whatever value is written directly to the DO Register
- **Fan Control** — driven by temperature thresholds, not by an alarm
- **Charge Cutoff / Discharge Cutoff** — a fixed set of protection alarms relevant to that phase
- **Aggregate outputs** — mirror the system-wide warning and/or protection state

Every multi-alarm function uses **OR** logic: the output activates if **any one** of its underlying alarms is active.

Active / Passive rule: every multi-alarm function (Charge Cutoff, Discharge Cutoff, All Warnings, All Protections, All Warnings + Protections) becomes **Active** the instant any one of its listed alarms is active, and only returns to **Passive** once **every** one of its listed alarms has cleared. An individual alarm clears according to its own alarm-set rules (see the Alarm Reference Document): the measured value must recover past the limit by that alarm's threshold margin. **A protection-type (PROT) alarm whose trigger counter (NOT) has reached its configured limit does not clear on its own** — it requires a manual operator reset, and any digital output tied to it stays Active until that reset is performed. Warning-type (WARN) alarms have no trigger counter and always clear automatically once their condition resolves.

2. Function Code Reference

Code	Function Name	Category	Description
0	User Configured	Basic	Firmware does not drive this output — Active when the operator/SCADA writes logic-high to the DO Register, Passive when it writes logic-low
1	Fan Control	Basic	Temperature-based control (see §6)
2	Charge Cutoff: OV+OT+UT+OC	Charge	Activates on any charge-related protection alarm (see §3)
3	Discharge Cutoff: UV+OT+UT+OC	Discharge	Activates on any discharge-related protection alarm (see §4)
4	All Warnings	Aggregate	Activates on any active warning alarm
5	All Protections	Aggregate	Activates on any active protection alarm
6	All Warnings + Protections	Aggregate	Activates on any active warning or protection alarm

Category key: Basic = single-purpose function — Charge/Discharge = phase-specific cutoff — Aggregate = mirrors overall alarm state

3. Charge Cutoff (Code 2)

Activates when **any** of the following alarms is active:

ID	Alarm Name
4	Pack Over Voltage Protection
6	Cell Over Voltage Protection
14	Charge Over Current Protection
18	Charge Over Temperature Protection
22	Charge Under Temperature Protection

Active: any one of the 5 alarms above is active. **Passive:** all 5 alarms above are inactive (subject to the [manual-reset caveat](#) for any that hit their NOT limit).

4. Discharge Cutoff (Code 3)

Activates when **any** of the following alarms is active:

ID	Alarm Name
10	Pack Under Voltage Protection
12	Cell Under Voltage Protection

ID	Alarm Name
16	Discharge Over Current Protection
20	Discharge Over Temperature Protection
24	Discharge Under Temperature Protection
61	Short Circuit Protection

Active: any one of the 6 alarms above is active. **Passive:** all 6 alarms above are inactive (subject to the [manual-reset caveat](#) for any that hit their NOT limit).

5. Aggregate Fault Outputs (Codes 4–6)

5.1 All Warnings (Code 4)

Activates when **any** of the following warning alarms is active:

ID	Alarm Name
5	Pack Over Voltage Warning
7	Cell Over Voltage Warning
11	Pack Under Voltage Warning
13	Cell Under Voltage Warning
15	Charge Over Current Warning
17	Discharge Over Current Warning
19	Charge Over Temperature Warning
21	Discharge Over Temperature Warning
23	Charge Under Temperature Warning
25	Discharge Under Temperature Warning
27	Low Capacity Warning
30	High Pressure Warning
32	Current Sensor Fault
49	Delta Temperature Warning
52	Current Sensor Reverse Connection Warning
53	Low Piezo Warning
57	LOG Write Warning
59	High Supply Warning

ID	Alarm Name
60	Low Supply Warning

Active: any one of the 19 warning alarms above is active. **Passive:** all 19 warning alarms above are inactive. WARN alarms always self-clear, so no manual reset can ever hold this output Active.

5.2 All Protections (Code 5)

Activates when **any** of the following protection alarms is active:

ID	Alarm Name
4	Pack Over Voltage Protection
6	Cell Over Voltage Protection
10	Pack Under Voltage Protection
12	Cell Under Voltage Protection
14	Charge Over Current Protection
16	Discharge Over Current Protection
18	Charge Over Temperature Protection
20	Discharge Over Temperature Protection
22	Charge Under Temperature Protection
24	Discharge Under Temperature Protection
26	Low Capacity Protection
31	LIN Sensor Fault
33	Temperature Sensor Fault
34	High H2 Protection
35	High CO Protection
47	Control Board Over Temperature Protection
61	Short Circuit Protection
62	Balance Break Protection

Note: ID 29 (Cell Unbalance Protection) is active in V1012 firmware but is **not** included in this aggregate — an active cell unbalance condition will not be reflected on a Code 5 output.

Active: any one of the 18 protection alarms above is active. **Passive:** all 18 protection alarms above are inactive **and** none of them is stuck waiting on a manual reset after hitting its NOT limit (see the [Active/Passive rule](#)).

5.3 All Warnings + Protections (Code 6)

Activates when **any** alarm listed in §5.1 or §5.2 is active — the logical OR of the two aggregates above (including the Code 5 exclusion noted there).

Active: any one of the 19 warning alarms in §5.1 or the 18 protection alarms in §5.2 is active. **Passive:** all 37 alarms across both lists are inactive, with the same manual-reset caveat as §5.2 for the protection side.

6. Fan Control (Code 1)

Fan control is driven by two configurable temperature thresholds, not by an alarm:

- **High limit:** the fan turn-on temperature threshold parameter
- **Low limit:** the fan turn-off temperature threshold parameter
- **Monitored sensors:** the sensor set depends on the BMU hardware model:
 - **BMU 500 mA model:** ambient temperature sensors 1 and 2
 - **BMU 100 mA model:** ambient NTC sensors 1 through 5
- **Behavior (checked in this order):**
 1. If **any** monitored sensor's reading exceeds the high limit, the output is **Active** (ON).
 2. Otherwise, if **any** monitored sensor's reading drops below the low limit, the output is **Passive** (OFF).
 3. Otherwise (every sensor sits between the two limits), the output **holds its previous state** — this is the hysteresis band.

Fan control runs independently of the alarm system — it is a standalone temperature-management feature, not tied to a fault ID.

7. Notes for Users

- **One function per output:** each digital output can be assigned exactly one function code.
 - **Only codes 0–6 are implemented:** the parameter accepts higher values, but firmware defines no behavior for them — entering a value above 6 leaves the output's prior configuration unchanged rather than selecting a new function.
 - **OR logic:** every multi-alarm function activates when **any one** of its listed alarms is active.
 - **Fan control is independent:** it runs on its own temperature thresholds, separate from the alarm system.
 - **All protective functions route through the alarm system:** the digital output only reflects an alarm's state; it does not evaluate conditions on its own.
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